

ROLL NO:250701573

EXPERIMENT NO:18 EXPERIMENT TO UNDERSTAND MATPLOTLIB LIBRARY USE
CASES IN DATA SCIENCE THROUGH VISUALISATION

PROGRAM:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline

data = pd.read_csv('Iris_Dataset_modified.csv')

data
data.info()
data.describe()
data.value_counts('variety')
sns.countplot(x='variety', data=data)

plt.show()

dummies = pd.get_dummies(data.variety)

FinalDataset = pd.concat(
    [pd.get_dummies(data.variety), data.iloc[:, [0,1,2,3]]],
    axis=1
)
```

```
FinalDataset.head()
```

```
sns.scatterplot(  
    x='SepalLengthCm',  
    y='SepalWidthCm',  
    hue='variety',  
    data=data  
)
```

```
sns.scatterplot(  
    x='PetalLengthCm',  
    y='PetalWidthCm',  
    hue='variety',  
    data=data  
)
```

```
plt.show()
```

```
plt.show()
```

```
sns.pairplot(data, hue='variety', height=3)
```

```
sns.FacetGrid(  
    data,  
    hue='variety',  
    height=5  
) .map(sns.histplot, 'PetalLengthCm').add_legend()
```

```
plt.show()
```

```
plt.show()
```

```
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(sns.histplot, 'PetalWidthCm').add_legend()
```

```
plt.show()

sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(sns.histplot, 'SepalLengthCm').add_legend()
```

```
plt.show()
```

OUTPUT:

Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety	
0	1	5.2	3.4	1.8	0.6	Iris-setosa
1	2	4.9	3.3	2.2	0.4	Iris-setosa
2	3	4.8	3.5	1.3	0.2	Iris-setosa
3	4	5.1	2.9	0.8	0.1	Iris-setosa
4	5	4.6	3.5	1.1	0.1	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	2.9	5.9	2.5	Iris-virginica
146	147	6.7	3.1	5.4	1.9	Iris-virginica
147	148	6.9	2.8	5.6	2.0	Iris-virginica
148	149	6.8	3.1	6.1	1.9	Iris-virginica
149	150	6.5	2.8	5.4	2.1	Iris-virginica

150 rows × 6 columns

```
<class 'pandas.DataFrame'>
```

RangeIndex: 150 entries, 0 to 149

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	Id	150 non-null	int64
1	SepalLengthCm	150 non-null	float64
2	SepalWidthCm	150 non-null	float64
3	PetalLengthCm	150 non-null	float64
4	PetalWidthCm	150 non-null	float64
5	variety	150 non-null	str

dtypes: float64(4), int64(1), str(1)

memory usage: 7.2 KB

Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.822667	3.091333	3.797333	1.223333
std	43.445368	0.742131	0.322935	1.796938	0.744683
min	1.000000	4.500000	2.400000	0.800000	0.100000
25%	38.250000	5.100000	2.800000	1.725000	0.400000
50%	75.500000	5.900000	3.100000	4.400000	1.400000
75%	112.750000	6.400000	3.300000	5.300000	1.900000
max	150.000000	7.300000	4.000000	7.000000	2.500000

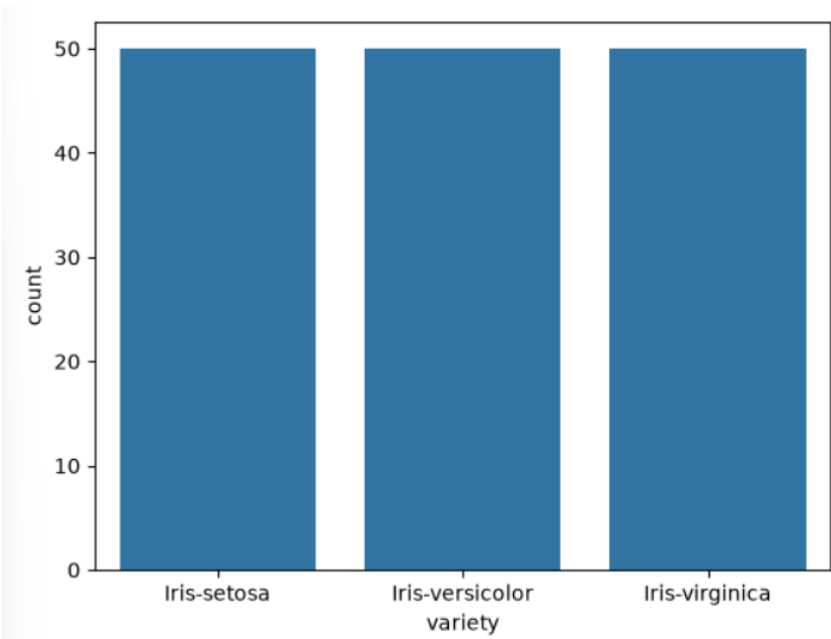
variety

Iris-setosa 50

Iris-versicolor 50

Iris-virginica 50

Name: count, dtype: int64



Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	
0	True	False	False	1	5.2	3.4	1.8
1	True	False	False	2	4.9	3.3	2.2
2	True	False	False	3	4.8	3.5	1.3
3	True	False	False	4	5.1	2.9	0.8
4	True	False	False	5	4.6	3.5	1.1

